Memory Banking in 8086

Q. What is Memory Banking in the 8086 microprocessor? Explain the roles of the A0 and BHE' signals in accessing Even and Odd memory banks for byte and word transfers.

Definition to Remember

***Memo*
*Bank***
8-bit byt

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1. Concept of Memory Banking

To support a 16-bit data bus while maintaining byte-level addressing, memory is split:

* **Ev
bus (**D

2. Role of A0 and BHE' Signals

The CPU uses two specific signals to select which bank to activate during a memory cycle:

* **A0

3. Byte and Word Access Scenarios

| BHE' | A0 | Selection | Transfer Type |

| :---: | :

4. Accessing a Word from an Odd Address (Unaligned Access)

If the CPU tries to read a 16-bit word starting at an odd address (e.g., `00001H`), it cannot do it in one cycle. It requires two clock cycles:

* **Cy

Must-Write Points (for 10 marks)

*1. **Mer*

Quick Recall

`Memor
in 1 cyc